CERTAIN-COAST — Text for the NATO SPS application form

Multi-Year Project · call 2027-1 · every field in the order of the official form · apply on https://natosps.grantplatform.com

How to use this document

• **This is the only document you need for the online form.** Go through it from top to bottom and copy each answer into the field with the same name on the platform. The other CERTAIN-COAST documents are background and are not uploaded.

• **Word limits are respected:** the green line under each long answer gives its length.

• **Yellow text** is information only you can give (names, dates, signatures, choices). Replace it before submitting.

• **Uploads and signatures** are listed in the checklist below; all uploads are PDF files.

• **Deadline:** 15 November 2026, 23:59 CET (call 2027-1, checked on nato.int on 6 November 2026). The call asks for full proposals in one round: fill every field now. Limits per field are in words; there is no page limit except one page per institution overview.

Checklist: uploads and signatures

Upload	Format	Who signs or prepares it	Status
SPS MYP Acceptance of Funds (NATO template)	PDF	Filled in: CERTAIN-COAST_SPS_MYP_Acceptance_of_Funds.docx; the NPD and a representative of the NPD's institution (Head of Institution or Financial Authority) sign; upload in tab 4	[to do]
SPS MYP Agreement for Joint Research (NATO template)	PDF	Filled in: CERTAIN-COAST_SPS_MYP_Agreement_for_Joint_Research.docx; all Co-Directors and a representative of each institution sign	[to do]
SPS MYP Detailed Budget workbook (NATO Excel template, tabs "Info" and "Detail")	Excel	Fill by hand (the form warns that using AI on this file may corrupt it and lead to automatic rejection), typing the values of CERTAIN-COAST_Budget_Workbook_Guide.docx cell by cell	[to do]
Project schedule with a Gantt chart	PDF	Ready: CERTAIN-COAST_Project_Schedule.pdf	ready
End-user commitments, all letters merged into one file	PDF	The Institute of Geosciences (IGJEO), Polytechnic University of Tirana: letter of support and end-user commitment on its letterhead, signed by Prof. Dr. Ylber Muceku (CERTAIN-COAST_Letter_IGJEO_EN_SQ.docx, page 1)	[to do]
Brief overview of participating institutions (at most one page each)	PDF	Ready: CERTAIN-COAST_Institutions_Overview.pdf (3 pages: UPT, Qatar University, IGJEO)	ready
Financial rules of the NPD's institution	Link	Ready: the UPT Statute in English (Chapter IV, Articles 24–25), https://upt.edu.al/wp-content/uploads/2023/02/Statuti-PDF-English.pdf. The UPT finance office confirmed on 10 November 2026 that UPT has no separate internal financial regulation and applies the national legal framework, including Instruction no. 10 of 12.07.2024 of the Ministry of Finance on foreign-funded programmes; these acts are named in the answer below	ready
Intellectual Property Rights agreement or waiver	PDF	Draft: CERTAIN-COAST_IPR_Agreement_Draft.docx; all Co-Directors, their institutions and the end-user sign; recommended now (listed in the 2027-1 call), allowed after approval	[to do]

Upload	Format	Who signs or prepares it	Status
Illustrative figures (optional, one PDF with a table of contents)	PDF	Ready: CERTAIN-COAST_Illustrative_Figures.pdf (6 pages, 12 figures)	ready
Additional files (optional): state of the art, technical annex, two notes, a report and the CVs of the two Project Directors	PDF	Ready: CERTAIN-COAST_Additional_Files.pdf (92 pages, the PDF of the Word file CERTAIN-COAST_Additional_Files.docx with the equations in native Word math; a cover, contents with live page numbers, what is new in each part and bookmarks: the state of the art, 15 pages, with 30 key papers of 2020–2026, what they have not done and what CERTAIN-COAST brings; the mathematical state of the art, 7 pages, on random forcing, boundaries, initial state and propagation of uncertainty; the technical annex, 10 pages, with methods, preparatory tests, gates and fallbacks and the nine new contributions; the technical note on the assembled system and the placement of the wells, 23 pages; the BEYOND-GO theory note on the inland boundary of the model, 16 pages; the SHOCK-GO implementation report on the earthquake in the aquifer, 14 pages; the CV of the NPD, 2 pages; the CV of the PPD, 2 pages, from the partner form of Qatar University) (build_additional_word.py)	Ready

Still missing before you can submit: [the signature and stamp of the Rector of UPT on the Acceptance of Funds and on the Agreement for Joint Research, which Qatar University signed on 5 November and the NPD on 10 November]; [the signed letter of support and end-user commitment of the Institute of Geosciences (IGJEO)]. The Rector's signatures confirm the NPD's institution; no decision of a department meeting is needed.

1. Start here

Project title

About 100 characters; understandable to non-experts; link to security and NATO.

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

100 characters

Project short title

80 characters; must match cell E2 ("Short Title") in the "Info" tab of the SPS MYP Detailed Budget workbook.

CERTAIN-COAST

Confirmation that you have read the Multi-Year Project Handbook

[Tick after reading the latest MYP Handbook.]

How did you hear about this opportunity?

[Your choice: word of mouth, journal, conference, online, SPS Information Day or other.]

Is the proposal a resubmission of a previous unsuccessful application?

[No] (unless you applied with this project before).

Did you use any generative AI tools in the preparation of your research proposal?

Yes. Parts of the text were drafted with an AI assistant and checked by the applicants. Answer honestly.

2. Eligibility

Are the Co-Directors (including NPD and PPD) affiliated with a governmental, academic, or other non-profit institution?

Yes. The NPD is employed by the Polytechnic University of Tirana and the PPD by Qatar University (Environmental Science Center), both public universities.

Do any of the Co-Directors (including NPD and PPD) hold other ongoing SPS grants?

No. The PPD has confirmed that he holds no ongoing SPS grant; [NPD: confirm] (an individual may hold only one SPS grant at a time).

Is the application developed jointly by one or more Co-Directors from NATO countries and one or more from eligible NATO partner countries?

Yes: the NPD in Albania (NATO member) and the PPD in Qatar (eligible NATO partner country, Istanbul Cooperation Initiative). Then click "Check Eligibility".

3. Activity Info

Primary Key Priority

Select one in the dropdown.

Environment, Climate Change and Security

Additional key priorities (optional, at most three)

Resilience; Critical Underwater Infrastructure; Innovation and Emerging Disruptive Technologies (EDTs)

What security challenge(s) would your project address?

500 words.

Three kinds of critical infrastructure of a NATO Ally stand on the same low Adriatic coastal plain of Albania, at two sites. At Durrës lie the wellfields that supply drinking water and the Port of Durrës, from which 93.6% of Albania's seaborne freight left in the second quarter of 2025 (INSTAT). Further south, near Fier, the Trans Adriatic Pipeline (TAP) leaves the shore for its 105 km crossing to Italy, carrying gas from Azerbaijan to Europe. The plain is built of soft sediments that settle slowly (Métois et al., 2020); at Durrës the ground is sinking close to active thrust faults, and the Mw 6.4 earthquake of 26 November 2019 caused liquefaction along the coast.

The assets face three linked hazards. Seawater intrudes into the aquifer and can make wells unusable for years. In an earthquake, saturated sand can liquefy under quays, foundations and the landfall. Storm surges and extreme sea levels flood low ground. Land subsidence and sea-level rise worsen all three at once, because they raise the sea and the water table together. Pumping moves the hazards in opposite directions: it draws seawater towards the wells but lowers the water table and so strengthens the ground against liquefaction. Managing one hazard alone can increase another.

Today these hazards are monitored by different institutions with different models, and no warning states how often it misses. For security decisions this is the critical gap: whoever must cut pumping, restrict a quay or inspect the landfall needs to know how far a warning can be trusted, and the national authorities need one consistent picture of water, port and energy after an earthquake or a storm.

The security challenge is therefore to keep three essential services of an Ally working, and to know in advance when they will stop being safe. The project serves three of NATO's seven baseline requirements for national resilience (resilient food and water resources, resilient civil transportation systems and resilient energy supplies) and the topic of particular interest of the 2027-1 call, "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". NATO's 2022 Strategic Concept calls climate change "a crisis and threat multiplier"; on this coast the multiplication is physical and measurable.

A second challenge is trust in machine-learned tools used in security decisions. CERTAIN-COAST uses artificial intelligence only with a certified error bound and a miss rate that is checked in operation, in line with NATO's Principles of Responsible Use of AI.

408 words (limit 500)

What technical challenge(s) would your project address, if any?

500 words.

Four technical problems stand between current science and a warning that institutions can act on.

1. The hazards are coupled but modelled apart. Seawater intrusion, aquifer compaction, liquefaction and coastal water levels depend on the same two variables, relative sea level and the water table, yet are computed by separate models. A pumping decision cannot be checked against all of them at once.

2. Relative sea level must be known at each asset, not only at the tide gauge. The land at Durrës moves vertically by different amounts from place to place, including the 2019 coseismic step; this motion must be measured with satellite radar (InSAR) and GNSS and carried into every hazard model with its uncertainty.

3. Physical models are too slow for warnings, and machine-learned copies come without a guarantee. Neural-operator surrogates make the thousands of daily runs possible, but a surrogate used for safety must state its error for each forecast.

4. No warning in this field states how often it misses. Recent statistical methods (online, regime-wise and extreme conformal prediction, and anytime-valid verification) can provide such guarantees, but they have not been applied to coupled coastal hazards driven by slow physics, with dependent alarms and few extreme events.

Six further problems have no published solution. There is no method to compute the set of pumping and dewatering decisions that keeps all the hazards within certified limits at once, nor the sea level or year at which this set disappears; tipping points exist for seawater intrusion alone, not for the coupled system. And monitoring networks are designed to reduce today's forecast errors; there is no criterion for how many years of safe operation a sensor buys. And routine pumping never shows how the aquifer responds near the onset of intrusion, which decides how long safe operation lasts; we found no pumping-test design that reveals it safely and values the test in years. And the source of rising chloride is often unknown: seawater, deep brine drawn up under the wells, a drought that concentrates the recharge, or irrigation return flow and wastewater. Each calls for a different action, and acting on the wrong one can make the salinity worse, yet no method lets water managers and their stakeholders decide before the source is known, with a risk that holds whatever it is. And every model of the aquifer is cut somewhere inland: a fixed head, a prescribed inflow or a general-head boundary does not respond to pumping or to the sea as the aquifer beyond does, and in tests this choice changed the fresh water lost to sea-level rise ninefold; no method learns the true boundary from data while keeping the model stable. And after an earthquake the pore-pressure step depends on each layer's stiffness; in tests the usual estimate overstated it 35 times in soft sand.

All of this must work on a real coast with incomplete records, under security restrictions on data, and in a form that the national authorities can use after the project.

499 words (limit 500)

What will the project deliver or develop by the end of the grant?

500 words; show how it advances the state of the art.

By Month 36 the project delivers:

1. A coastal motion atlas from Durrës to Fier and relative sea level at each asset, from Sentinel-1 InSAR, reflectors and GNSS (D1.1).

2. A monitoring network chosen for the warnings, in the project's own boreholes on public land (D4.2): piezometers with conductivity loggers, a barometer and sealed hub channels that correct them, a rain gauge, GNSS, seismic nodes and, where access allows, fibre-optic sensing.

3. One calibrated model that couples groundwater flow and salt, aquifer compaction, liquefaction under a moving water table and coastal water levels, its inland boundary learned from the heads (D2.1), and a fast neural-operator copy with a certified error bound (D2.2).

4. Certified alarms for seawater intrusion, liquefaction and extreme water levels whose miss rate is bounded in advance (indicatively 5–10%), with a joint guarantee for the three assets, back-tested since 2014 under a pre-registered protocol (D3.1, D3.2).

5. Decisions for the wellfields: the effect of each wellfield's pumping on salinity with the hidden drought demand removed; the salinity under drier recharge; the certified safe operating window and the year it closes; safe pumping tests planned to sharpen it; a pumping rule; and pumping safe while the source of the chloride is unknown.

6. Twelve months of live bulletins with the end-user, every outcome logged and the miss rates verified (D5.1); operating procedures and a policy brief (D6.2); the model and the alarm algorithm, built by the NPD, released as open-source MATLAB code (https://github.com/dulianzeqiraj/certain-coast; https://doi.org/10.5281/zenodo.24148762) with open data and a DOI (D6.1).

How this advances the state of the art. The alarms apply conformal and anytime-valid methods of 2023–2026 to coupled coastal hazards. Nine new contributions were tested before the project. (i) Guarantees for quantities nobody measures: inputs and boundaries known only through rain, with bands valid when the simulator or rain product is wrong. (ii) The value of a sensor as the guaranteed band it removes and the years of safe operation it adds. (iii) Pumping effects free of hidden confounding, read from the heads. (iv) The certified safe operating window and safe pumping tests chosen for the years they add. (v) A network that corrects its own sensors for air pressure and logger drift. (vi) Warnings between samples: a certified band for the chance that chloride crosses its limit at any time in the season (34 times the exceedance on sampling dates in tests). (vii) Decisions before the source of the salt is known: pumping safe for every source the tracers cannot exclude (in tests ≤6.9% risk for a 10% budget; assuming seawater intrusion: 98%). (viii) A boundary that learns the aquifer beyond the model: exact for any linear aquifer, stable by construction, learned from two piezometers (6 mm against 372 mm on an unseen drought in tests). (ix) Which layer feels an earthquake, and its step as a free pumping test (in tests the usual estimate overstated the soft-sand step 35 times). The integrated system reaches TRL 6 on the project's network at Durrës.

498 words (limit 500)

How will the project reach the planned end state?

500 words; methodology, approach and how the teams collaborate.

Seven work packages run over 36 months in three phases, separated by evidence gates with thresholds fixed in advance.

Observe and measure (Months 1–12). WP4 designs the network before drilling or buying it: each candidate sensor (conductivity loggers, a rain gauge, GNSS, seismic nodes, a barometer and hub reference channels) is ranked by the guaranteed warning band it removes and the years of safe operation it adds per euro; the network is chosen against the worst plausible sensor error, with a certified optimality gap (D4.1, Month 6). WP1 obtains vertical land motion and relative sea level at each asset from Sentinel-1 InSAR since 2014 and GNSS, and the head limit for pumping from aquifer compaction.

Model and certify (Months 4–30). WP2 builds one coupled model in stages (flow and salt, then compaction and liquefaction). The inputs and boundaries nobody measures are reconstructed from rain and heads by a generative prior with an ensemble update. Their bands, calibrated on simulated aquifers and measured anchor periods, stay valid if the simulator is wrong or the climate drifts. The inland boundary is learned from two piezometers as passive storage compartments. Gate MS3 (Month 18) requires held-out heads, chloride, InSAR and seismic velocity inside the 90% intervals. A neural-operator copy with a certified error bound then serves the warnings. WP3 agrees thresholds and tolerable miss rates with the end-user, builds the alarms (crossings between samples included), back-tests them since 2014 under a protocol registered before the results; gates MS4 (Month 22) and MS5 (Month 27) decide live operation and confirm the miss rates.

Operate, verify and hand over (Months 22–36). WP5 runs the system live for twelve months, logs every bulletin with its outcome, verifies the miss rates with anytime-valid tests and runs a post-event protocol. It estimates what each wellfield's pumping does to the salinity with the hidden drought demand removed, reading the recharge from the heads under a calibrated check. While the source of the chloride is uncertain, tracers chosen for the decisions they change keep only the sources the data cannot exclude, updated monthly, and the pumping stays safe for all of them. It computes the salinity under drier recharge, the certified safe operating window, its closing year and safe pumping tests that would sharpen it, and prepares a pumping rule for the water-use permits. WP6 trains early-career researchers and the end-user's staff, releases the open-source code and transfers the method to a partner coast. WP7 manages data access, security classification, risks and reporting.

Collaboration. The NPD builds the coupled model and the alarm algorithm and releases them as open-source MATLAB/Octave code; the Albanian team leads WP1–WP5 and WP7; Qatar University leads WP6, a transfer study on the east coast of Qatar and a user-side review of the bulletins. Early-career researchers of both countries work across work packages, with two-month exchanges. The end-user agrees the thresholds and reviews the bulletins; a steering committee with it meets every six months, and each gate has a named fallback.

495 words (limit 500)

May your proposal raise any ethical issue?

Yes, limited: personal data of workshop and training participants, and security-sensitive data about critical infrastructure. If a text box appears, use:

CERTAIN-COAST involves no animals, human tissue or clinical research. Its ethical issues are people and their data, security-sensitive information about critical infrastructure, machine learning in safety decisions, field work and the responsibility of issuing warnings. Participants in workshops, interviews and tests of the bulletins take part voluntarily with informed consent, after institutional ethics approval; personal data are minimised, pseudonymised and protected under Albania's Law No. 124/2024, aligned with the GDPR. Asset-level vulnerability results are shared only with the end-users and the competent authorities through a release gate. The machine-learned surrogate follows NATO's six principles of responsible AI: every forecast carries an error bound, every bulletin is logged and reproducible, and the users keep the decision. Each bulletin states its certified miss rate, the miss-false-alarm trade-off is agreed with the users in advance, public warnings remain with the authorities, and the back-testing protocol is frozen before results are seen. An independent ethics advisor and a yearly review oversee these measures.

State-of-the-art: Related Projects / Duplication

Related research outside the project team (2020–2026):

- Coastal groundwater under climate change and pumping. Rising sea level lifts the water table and extends flooding inland (Befus et al., 2020); 250,000 US coastal wells show where heads lie below sea level (Jasechko et al., 2020); by 2100 recharge sets the severity of intrusion and sea level its prevalence (Adams et al., 2024); ensemble pumping optimisation handles parameter and climate uncertainty (Coulon et al., 2024); machine-learned surrogates predict intrusion fast (Jiang, Liu and Dwivedi, 2025). These give scenarios without guarantees and treat pumping as known and exogenous.
- Inputs and boundaries from sparse data. Ensemble and diffusion-based inversion (Zheng et al., 2025; Yao et al., 2025), satellite and machine-learning recharge (Soylu, Entekhabi and Bras, 2026) and evaluations of gridded rain products (Abbas et al., 2026) estimate what is not measured, without a calibrated guarantee and without carrying the rain-product errors into the boundaries.
- Guarantees for forecasts and surrogates. Conformal risk control (Angelopoulos et al., 2024), online calibration under shift (Gibbs and Candès, 2024; Feldman et al., 2023), robustness to distribution shift (Aolaritei et al., 2025), conformal neural operators (Ma et al., 2024), extreme conformal prediction (Pasche, Lam and Engelke, 2026), the expected size of conformal sets (Dhillon, Deligiannidis and Rainforth, 2024) and anytime-valid inference (Ramdas et al., 2023) all calibrate on observed

outcomes; none covers a boundary that is never observed, asks how narrow such a guarantee can be, or how long it holds under drift.

• Monitoring design. Bayesian optimal design places sensors for PDE-governed inverse problems (Alexanderian, 2021; Wu, Chen and Ghattas, 2023); greedy selection is near-optimal for submodular criteria (Krause, Singh and Guestrin, 2008), which correlated errors break (Liu et al., 2016; Uciński, 2020; Attia and Constantinescu, 2022); robust designs maximise the worst case over an uncertain error model (Attia, Leyffer and Munson, 2023; Chowdhary, Attia and Alexanderian, 2025); recent work places several sensor types (Maio et al., 2026), designs tsunami warning networks (Venkat, Henneking and Ghattas, 2026) or ranks wells by predictive skill (Meray et al., 2022). None values a sensor by the guaranteed band it removes or the years of safe operation it adds, or certifies a robust network's distance from the best one; all take correlated errors as given, although air pressure and logger drift are routinely removed with a nearby barometer (Rasmussen and Crawford, 1997; Toll and Rasmussen, 2007), outside the design.

• Forecasting from irregular records and robust bounds. Neural jump ODEs learn the optimal online forecast from irregular, incomplete records (Herrera, Krach and Teichmann, 2021; Krach, Nübel and Teichmann, 2026); robust finance bounds a path-dependent price over all models that fit the liquid prices (Hobson, 1998; Beiglböck, Henry-Labordère and Penkner, 2013), computed with neural SDEs (Gierjatowicz et al., 2020). Neither gives a finite-record guarantee or names the observation that would narrow the bound. Yet sampling pins the marginal law of chloride, not its time scale, so a crossing of the limit between samples is bounded only from below (Slepian, 1962).

• Hidden confounding over time. The Time Series Deconfounder infers substitutes for hidden confounders from many treatments (Bica, Alaa and van der Schaar, 2020); proxy methods (Mastouri et al., 2021), synthetic interventions (Agarwal, Shah and Shen, 2026), conformal intervals for individual effects (Alaa, Ahmad and van der Laan, 2023), policy counterfactuals (McKay and Wolf, 2023), causal dose curves (Singh, Xu and Gretton, 2024) and adjoint attribution in the ocean (Fukumori, Wang and Fenty, 2021) complete the picture. Substitutes built from the treatments cannot see the fresh news of a confounder such as a heat wave that makes every farmer pump, and none uses a physical measurement of the confounding path.

• Decisions when the cause of the salt is unknown. Drought propagates into groundwater with measurable lags (Farahmand, Reager and Madani, 2021; Entekhabi, 2023); salinisation acts as an aridity enhancer (Perri et al., 2020); water planning is framed as a control problem under deep uncertainty (Herman et al., 2020); drinking-water investment on a salinised coast trades total access against access for the poorest (Roman et al., 2021); scenario-based robust optimisation protects against discrete uncertainty (Wang, Aydemir and Jacquillat, 2024); legitimacy decides whether groundwater users accept limits (Jedd et al., 2024). None decides before the source of rising chloride is known with a risk stated for every source, although e-values (Vovk and Wang, 2021) and universal inference (Wasserman, Ramdas and Balakrishnan, 2020) give the tools.

• The boundary where a model is cut. Groundwater models end inland at a fixed head, a prescribed flux or a general-head boundary (Langevin et al., 2017), and this choice decides how intrusion responds to sea-level rise (Werner and Simmons, 2009; Michael, Russoniello and Byron, 2013). Exact boundaries built from sums of exponentials are classical for waves (Engquist and Majda, 1977; Alpert, Greengard and Hagstrom, 2000); passive rational models of an external network are standard in power systems (Gustavsen and Semlyen, 2001); head time series are modelled with positive impulse responses (von Asmuth, Bierkens and Maas, 2002; Bakker and Schaars, 2019). None learns the boundary of a groundwater model from data with its stability and identifiability guaranteed.

• Earthquakes in coastal aquifers. Fault models give the static strain near the source (Okada, 1985); undrained volumetric change explains abrupt water-level changes near the source, and earthquake-enhanced permeability the sustained ones farther away (Wang and Chia, 2008). The 2019 Durrës earthquake has four published source models (Ganas et al., 2020; Govorčin et al., 2020; Matraku et al., 2023; van der Heiden et al., 2026). CPT-based triggering (Boulanger and Idriss, 2016) and the liquefaction potential index (Maurer et al., 2014) take the water table as given. None decides which

layer of a soft coastal plain feels the static step, or reads the leakage of a confined aquifer from the relaxation of that step.

• Vertical land motion and coastal exposure from InSAR and GNSS (Ohenhen et al., 2023), and slow settling of the Albanian coastal plain near Fier (Métois et al., 2020); the 2019 Durrës earthquake and its liquefaction (Mavroulis, Lekkas and Carydis, 2021).

What none of these does. No study combines intrusion, liquefaction and extreme water levels in one model with warnings whose miss rate is stated in advance; guarantees the inputs known only through rain when the simulator is wrong; values a sensor by the guaranteed band it removes and the years it adds; places the reference channels that correct the sensors; certifies crossings between samples; reads the hidden drought demand from the heads; decides how to pump before the source of the salt is known; learns the inland boundary of the model from data without putting its stability at risk; tells which layer an earthquake loads; or computes when a safe way of operating the assets ends. A final check in bibliographic databases will be made before submission.

Related activities of the project participants: the NPD leads the project of the National Agency for Scientific Research and Innovation (AKKSHI) "Modelling, simulation and optimisation of formation treatment methods to maximise the net profit of the massive carbonate reservoir of the Amonicë oil field" (2025–, with Albpetrol; [budget]), which builds a 3D reservoir model and validates it against field data. The Qatar team runs Qatar University projects on shoreline change between Al Arish and Al Ghariyah (2025–2027) and on wind and wave modelling for the Exclusive Economic Zone of Qatar (2025–2027). The preparatory technical work of the NPD's team in 2026 (eleven reviews with tests, a calibrated chloride model of the Fushë-Kuqe aquifer) is the starting point of the project, not a separately funded activity.

Other activities at the site known to the applicants: [list any national, EU or donor projects on subsidence, the port or the water supply of Durrës, or write "none known"].

What makes CERTAIN-COAST stand out: one model for three hazards at three critical assets, warnings and pumping decisions whose error is stated in advance and verified in operation, and a certified planning horizon for the institutions that manage the assets.

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Funding requested from NATO SPS (in EUR)

Must equal cell G15 of the "Summary" tab of the SPS MYP Detailed Budget workbook.

382,500

4. NPD & PPD

Fill these fields directly on the platform, once for the NATO-country Project Director (NPD) and once for the partner-country Project Director (PPD). For the PPD, Part 1 of the revised partner form of 9 November (qatar/1_nga_Katari/2026-11-09_Partner_Form_RISHIKUAR_ekipi_i_ri.docx) uses exactly these fields: copy it field by field; his personal and contact data are only there.

Fields	NPD (Dulian Zeqiraj)	PPD (Jassim Al-Khayat)
First name, family name, date of birth, sex, title	[personal data]	Jassim Al-Khayat, Prof.; date of birth and sex: [from the partner form]
Nationality (and other nationalities)	Albanian	Qatari
Telephone, mobile, personal and work e-mail; online profile (optional)	[contact data]	[from the partner form]
Current job title; name, country and postal address of the institution	Associate Professor; Faculty of Geology and Mining, Polytechnic University of Tirana, Albania; [postal address]	Research Professor and Manager of the research vessel Janan; Environmental Science Center, Qatar University, P.O. Box 2713, Doha, Qatar
Employment history (three rows)	2008–: Faculty of Geology and Mining, UPT (lecturer, then associate professor); 2006–2008: Albanian Geological Survey; 1996–2006: Alpetrol (as in the CV of the NPD)	Feb 2021–: Research Professor and Manager of RV Janan, ESC, Qatar University; Oct 2007–Feb 2021: Associate Professor and Coordinator of the Marine Sciences Program, Qatar University; Sep 1996–Oct 2007: Assistant, then Associate Professor and Head of the Marine Sciences Department, Qatar University

Fields	NPD (Dulian Zeqiraj)	PPD (Jassim Al-Khayat)
Education (three rows)	[doctorate: year, university]; 1991–1996: university diploma, Faculty of Geology and Mining, University of Tirana	1993–1996: PhD Marine Biology, University of Wales, UK; 1991–1992: MSc Marine Biology and Fisheries, University of Aberdeen, UK; 1979–1983: BSc Marine Science/Geology, Qatar University
Publications and honours (optional, up to three each)	The first three publications of the CV of the NPD	The three publications and two honours of Part 1 of the partner form (Outstanding Teaching Award, Qatar University, 2015; National Incentive Award for Research, Qatar, 2026)
Research focus and expertise (500 words)	[how the research of the NPD relates to the project: seawater intrusion, monitoring design, stochastic flow and transport]	The text of Part 1 of the partner form (about 470 words)
Other grants held (optional)	2025–: AKKSHI, project leader, "Modelling, simulation and optimisation of formation treatment methods to maximise the net profit of the massive carbonate reservoir of the Amonicë oil field", with Albpetrol; [budget]; [results so far]	None given
SPS Acceptance of Funds (NPD only)	Upload the signed PDF of CERTAIN-COAST_SPS_MYP_Acceptance_of_Funds.docx (official NATO template, filled in): signature of the NPD and of a representative of the NPD's institution	—

5. Co-Directors (optional)

Not required. If you add a Co-Director, for example [an additional NATO-country co-director], they must be at a different institution and have their own budget in the Excel workbook. Colleagues of the NPD's own faculty take part as participants, not as Co-Directors. Otherwise skip this tab.

6. Project Participants

Roles of Co-Directors and Participants (add rows as needed; all must be nationals of NATO or eligible partner countries):

First name	Family name	Affiliation	Position	Nationality	Task(s)	% time
Dulian	Zeqiraj	Faculty of Geology and Mining, Polytechnic University of Tirana	NATO-country Project Director	Albanian	Coordination; builds the coupled model (WP2) and the alarm algorithm (WP3) and writes the open-source code (D6.1); WP1 coastal motion; WP4 monitoring network; WP5 live operation; WP7 management	60%
Jassim	Al-Khayat	Environmental Science Center, Qatar University	Partner-country Project Director (Prof.)	Qatari	WP6 training, transfer and dissemination; transfer study on the east coast of Qatar (site, local knowledge, public data); regional workshop; user-side review of the bulletins	20%
Fares	AlMomani	Environmental Science Center, Qatar University	Research Professor; Acting Director of the Center	Canadian	Water quality, water resource management and groundwater protection at the transfer site (T6.4)	5%
Oguz	Yigiterhan	Environmental Science Center, Qatar University	Research Associate Professor	Turkish	Sampling of seawater, sediment, biota and aeolian dust; methods, data curation and dissemination (WP6)	5%
Bruno	Welter Giraldes	Environmental Science Center, Qatar University	Research Associate Professor	Portuguese	Data processing and management, data curation and publications (WP6)	5%
Arjan	Beqiraj	Faculty of Geology and Mining, Polytechnic University of Tirana	Senior researcher (Prof. Dr.)	[Albanian]	Hydrogeology and groundwater chemistry: tracer campaign and sources of the chloride (T4.1, T5.3); aquifer properties for the coupled model (T2.3)	30%

First name	Family name	Affiliation	Position	Nationality	Task(s)	% time
Bardhosh	Seitaj	Faculty of Geology and Mining, Polytechnic University of Tirana	Researcher (Dr.); Head of the Department of Energy Resources	[Albanian]	Drilling and installation of the piezometers (T4.2); pumping tests (T5.3)	20%
Esmeralda	Zeqo	Faculty of Geology and Mining, Polytechnic University of Tirana	Researcher (Dr.), petroleum drilling engineer, Department of Energy Resources	[Albanian]	Assists with the design of the boreholes and the quality control of drilling, casing and sealing of the piezometers (T4.2); well data for the model (T2.1)	20%
Aurela	Jahja	Faculty of Geology and Mining, Polytechnic University of Tirana	Early-career researcher (MSc, geochemist; stipend)	[Albanian]	Sampling, chloride and major-ion analyses, tracers (Br/Cl, water isotopes) and their interpretation (T4.1, T5.3)	30%
Sonata	Pregja	Faculty of Geology and Mining, Polytechnic University of Tirana	Laboratory technician	[Albanian]	Laboratory analyses of chloride and major ions (T4.1, T5.3)	10%
Armida	Brakaj	Faculty of Mathematical Engineering and Physics Engineering, Polytechnic University of Tirana	Early-career researcher (MSc, mathematician ; stipend)	[Albanian]	Mathematical methods: certified error bounds of the fast model and alarm margins (WP2, WP3)	30%
Elvana	Ndreca	Rectorate, Polytechnic University of Tirana	Financial officer	[Albanian]	Financial management of the grant at the NPD institution: project account, payments and procurement, Financial Record and financial reports to the SPS Office (WP7)	10%
To be recruited	(2 people)	Qatar University	Early-career researchers (stipends)	[NATO or partner]	Transfer study and data (WP6); exchanges in Albania	[50%]

7. Project plan

Project abstract

200 words; understandable to non-experts.

On Albania's Adriatic coast, the drinking-water wells and port of Durrës and the landfall of a gas pipeline to Italy near Fier stand on low ground, which at Durrës is sinking, close to active faults. Three dangers threaten them: seawater entering the wells, ground that turns almost liquid in an earthquake, and the sea flooding the land in storms. The dangers are linked, because sinking land and rising seas worsen all three and pumping that protects against one can increase another, but today they are watched separately and no warning says how often it can be wrong.

CERTAIN-COAST will watch them together. Satellites, GNSS and new ground sensors will feed one computer model of the coast, copied by a fast artificial-intelligence model whose error is stated for every forecast. Each warning will state in advance how often it may miss a dangerous event, and this will be checked in twelve months of live operation with the Institute of Geosciences (IGJEO), which monitors earthquakes, weather and waters and will use the warnings. The project will show how many years a safe way of operating the three assets remains, and plan safe pumping tests that would sharpen that estimate.

197 words (limit 200)

Security relevance and real-world applications

Link to security. The project protects three essential services of a NATO Ally (drinking water, a port that handles most of the country's seaborne freight, and the landfall of the Trans Adriatic Pipeline near Fier) and serves three of NATO's seven baseline requirements for national resilience: resilient food and water resources, resilient civil transportation systems and resilient energy supplies, which also underpin civil support to the military. It falls under the 2027-1 topic "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience": the TAP landfall is where an undersea gas pipeline between two Allies, Albania and Italy, leaves the shore.

Use cases:

- The Institute of Geosciences (IGJEO) of the Polytechnic University of Tirana, the end-user, which monitors earthquakes, weather and waters in Albania: receives the bulletins for the wellfields, the port and the landfall during the live operation, agrees the warning thresholds for earthquakes and extreme sea levels, reviews the bulletins, shares the data of its stations and can use the methods in its own monitoring; staff time and data in kind.
- The authorities that issue water-use permits and the water supply: a chloride early warning and a pumping and drought rule, offered as a policy brief, so that pumping at the wellfields can be cut or shifted before chloride reaches drinking-water wells, safely even while the source of the chloride is unknown.
- The Port of Durrës and the TAP landfall near Fier: warnings of ground motion, liquefaction and storm surges, computed from satellite radar, GNSS and public data, to set inspection priorities after an earthquake and prepare for surges (hours to days ahead); no data or access from the asset owners is needed.
- National authorities and resilience planners: the certified year from which no operating plan keeps the three assets safe together, and the sensors and measures that delay it most.

Each bulletin states its certified miss rate, so decision-makers know how far it can be trusted. The method uses free satellite data and transfers to other subsiding coasts of Allies and partners with wellfields, ports and energy landfalls.

Current status

The approach has been tested in preparatory work by the NPD's team (2026), on synthetic coastal aquifers and on the calibrated chloride model of the Fushë-Kuqe aquifer, which supplies Durrës:

- Boundaries known only through rain: on 500 simulated aquifers with four loggers, the raw ensemble missed 17–20% of the weeks; the calibrated bands met the 10% target (0.095–0.100).
- A wrong simulator: bands calibrated on simulations broke under a 35-day vadose delay, shifted parameters or a rain bias (risk 0.11–0.83); calibrated on 10–40 measured anchors they stayed at or below 10% under every law tested. At least 9 anchors are needed for an average guarantee and 22 for a 90% certificate.
- Drift: horizon-matched bands kept the risk at 0.003–0.082 up to 20 years ahead under every drift tested; with 30 anchor years a band exists only up to 17 years ahead.
- The rain product and the floor: a local gauge with one logger identified the recharge coefficient better (error 0.045) than the product with eight loggers (0.108); with 20 anchors the guaranteed band was 1.34–1.54 times the narrowest possible.
- Value of a sensor: a tide gauge that a standard ensemble said would narrow the recharge band by 15% widened the guaranteed band by 16–21%; the sign of each guaranteed value was predicted for 36 of 36 instrument–goal pairs.
- Counterfactual salinity at Fushë-Kuqe (calibrated model): with 40% less recharge the shallow area above 250 mg/L chloride would grow from 31 to 65 km² (effect +33 km², conformal band 15–49 km²), with coverage 0.887–0.900 on 300 simulated test sets.
- Pumping and hidden drought demand: in a test bed with eight wells, the summed effect of pumping on salinity (true value 0.94) was overstated by 0.59 without adjustment, by 0.56 with a substitute built

from the pumping history (as in the Time Series Deconfounder) and by 0.00 with the recharge read from the heads; the physics check had size 6% and power 99–100%.

• Alarms and operating window: an alarm with an extreme-aware margin kept the share of seasons above the chloride limit at 4–9% across storylines, where a static alarm allowed 13–21%; the certified safe operating window closed after 38 years, while plans that watch one hazard believed they had 43–97 years; safe pumping tests chosen for the years they add raised the certified closing year from 10.8 to 31.9 years (true value 45) with chloride below 123 mg/L in 95% of runs; the anytime-valid test wrongly flagged a correctly calibrated alarm in 2.2% of runs, against 68% for re-checking a 95% bound after every event.

• Sensors that interfere: a 12-sensor network with a barometer and hub reference channels kept 52% more worst-case information than the classical design, certified within 18% of the best; ignoring the interference let 90% intervals cover only 49–62% of cases, the correction 88–93%.

• Crossings between samples: chloride with an 8-day time scale crossed 250 mg/L within a season with probability 0.78, 34 times the exceedance on a sampling date; quarterly samples left this anywhere in 0.18–1, a daily logger in 0.76–0.80; a conformal margin restored 90.3% coverage for skewed chloride (74% without).

• Unknown source of the chloride: four rival sources (seawater, deep brine, drought, return flow) fitted today's chloride equally well; acting on the wrong one let the blend exceed 250 mg/L with probability up to 100% (98% when seawater intrusion was assumed); keeping only the sources the tracers could not exclude, and protecting against all of them, held every source at or below 6.9% against a 10% budget, at €3.31 million against €4.16 million for protecting against all four; monthly updates dropped the true source in 1.2–1.5% of cases, against about 22% for an ordinary test repeated every month.

• Boundary of the model: on a 10 km test aquifer cut at 2 km, a boundary of six positive storage compartments reproduced the head at the cut to 7.8 mm over 20 years, against 148 mm for the best calibrated general-head boundary; learned from two piezometers while the pumping varied, it predicted an unseen drought with doubled pumping to 6 mm (general-head boundary: 372 mm); under a 0.5 m sea-level rise it lost 160 m³ of fresh water per metre of coast against 154 in the full aquifer, where a prescribed inflow gave 49 and a fixed head 462; a negative conductance that was stable at 2 km grew without bound at 6 km.

• Earthquake in the aquifer: with four published source models of the 2019 earthquake, the static pore-pressure step was 7–12 cm of head in soft surface sand and 0.6–1.0 m in a stiffer confined aquifer, where the usual estimate gives 35 times the soft-sand value; for liquefaction the water table of the day dominates (in an example profile the liquefaction potential index fell from 20.2 to 9.3 as the water table deepened from 0.8 to 2.6 m). After a 0.3 m step, 20 days of logger data pinned the leakage time of the confined aquifer 5.8 times more sharply than a year of seasonal heads; a hindcast of the 2019 liquefaction map had a power below 50% even with 480 sites. At a synthetic port the window between liquefaction and intrusion closed at 0.29 m of sea-level rise for a design acceleration of 0.20 g, and the chance of an index above 5 if the earthquake struck that day moved between 0.16 and 0.26 over the year.

These are tests on synthetic coasts and on the existing chloride model with illustrative values, not results at Durrës; the project calibrates and verifies them on site.

Starting data: Sentinel-1 radar images since 2014, tide-gauge records, the groundwater records behind the existing chloride model of Fushë-Kuqe, published groundwater monitoring reports, and the mapped liquefaction of the 2019 earthquake (Mavroulis et al., 2021).

Team: the NPD works on seawater intrusion and the design of monitoring networks (Zeqiraj, 2026, Journal of Hydrology), Bayesian inference in alluvial aquifers (Zeqiraj and Beqiraj, 2026, Journal of Contaminant Hydrology) and stochastic modelling of flow and transport, and leads an AKKSHI project with Albpetrol; Prof. Beqiraj brings hydrogeology and groundwater chemistry; the team of the Environmental Science Center of Qatar University works on coastal ecology, shoreline change, waves, coastal water quality and the oceanography of the Arabian Gulf (Backer et al., 2026; Al-Ansari et al., 2022).

Impact

500 words; quantitative where possible.

If the project succeeds, the following will change at Durrës, at the TAP landfall near Fier and beyond:

- Three hazards covered by one warning with a stated miss rate, where today there is none: seawater intrusion at the wellfields, liquefaction at the port and the landfall, and extreme water levels at all three.
- Each alarm misses dangerous events no more often than its certified rate (indicatively 5–10%), shown by back-testing over the record since 2014 and checked in twelve months of live operation; the joint miss rate for the three assets is bounded by the sum of the rates.
- Each user knows in advance the minimum value of the warning for its own decisions, computed from the certified miss rate and the cost-loss ratio agreed with it.
- The wellfields that supply Durrës get a certified safe operating window and a plan of safe pumping tests that would sharpen it, which lowers the risk of salinising wells that take years to recover and avoids new permanent subsidence; planners know when the window closes and which measures extend it.
- The Institute of Geosciences (IGJEO), the end-user, gains warnings with a stated miss rate for the coast it monitors, data from the project's network, trained staff and procedures, and no procurement, ownership or maintenance obligations.
- The authorities that issue water-use permits and the water users can act before the source of rising chloride is known: tracers are bought for the decision they change, the plausible sources and the pumping in force are published every month with a risk that holds whatever the source, and a ledger shows who bears the cost.
- Models whose inland edge responds as the real aquifer does, learned from piezometers: sea-level rise and new pumping are then neither underestimated by a prescribed inflow nor overstated by a fixed head.
- After an earthquake, the authorities know which layers' pressure rose, and the monitoring wells measure the aquifer for free.
- At least 15 professionals trained, including at least four staff of the end-user; at least six early-career researchers trained in the NATO and partner countries.
- A monitoring network of piezometers, GNSS, seismic stations and corner reflectors that stays in Albania with the Polytechnic University of Tirana, with its running cost of about €10,900 per year set out in an agreement.
- Open-source code of the model and the alarm algorithm and open data with a DOI, installed and reproduced by a group outside the consortium; at least six open-access papers; an operating procedure agreed with the end-user; a policy brief for the national authorities.
- The method tested on the east coast of Qatar and ready for other subsiding coasts of the Adriatic and the Mediterranean.

For NATO, the result is a tested capability for three baseline requirements of resilience at an Allied site, and a worked example of artificial intelligence used in a security decision chain with testing, evaluation, verification and validation built in.

494 words (limit 500)

Technology Readiness Level (TRL)

TRL 6: Technology demonstrated in relevant environment

Technology Readiness Level (TRL) description

Start: TRL 3. Each component has been shown in preparatory synthetic tests (alarm margins, safe operating window, verification, guaranteed reconstruction from rain, value of a sensor, pumping effects with the hidden confounding removed, counterfactual salinity on the calibrated chloride model of Fushë-Kuqe, a network that corrects its own sensors, warnings for crossings between samples, decisions

before the source of the chloride is known, an inland boundary learned from piezometers, the pore-pressure step of each layer after an earthquake), and published methods exist for each element.

End: TRL 6, as the 2027-1 call asks (results at TRL 3–6). The integrated system (network, coupled model, certified surrogate, alarms, pumping rule and safe operating window) is demonstrated in its relevant environment, the project's monitoring network on the coast of Durrës and Fushë-Kuqe, through twelve months of live operation with the Institute of Geosciences (IGJEO) as end-user; at the port and the TAP landfall the warnings run on satellite, GNSS and public data. Seismic velocity as a coastal water-table sensor and the pumping rule are demonstrated at the site but not yet run for a full year; the safe pumping tests are planned and checked on the model.

Data exploitation plan

Data required and existing sources: Copernicus Sentinel-1 radar images since 2014 (free and open); GNSS records; the Durrës tide-gauge record of the State Authority for Geospatial Information (ASIG); groundwater level and chemistry records behind the existing chloride model of Fushë-Kuqe and published groundwater monitoring reports; abstraction records where their holders share them; published geotechnical data and the mapped liquefaction of 2019; records of existing national seismic stations; meteorological and wave data.

New data: continuous heads, conductivity and temperature in new and existing wells; GNSS and corner-reflector time series at the assets; continuous seismic noise from stations and temporary nodes, and fibre-optic sensing where access is agreed; laboratory chloride and major ions; the bulletin log with outcomes.

Collection and management: a data management plan (D7.1, Month 3) following the FAIR principles; standard formats (miniSEED and StationXML for seismic data, NetCDF with CF conventions for time series) and DataCite metadata; storage on the partners' servers with backup; open release of derived products and software with DOIs on Zenodo. The code is public from the start, under the Apache License 2.0: https://github.com/dulianzeqiraj/certain-coast, archived on Zenodo with the DOI https://doi.org/10.5281/zenodo.24148762 (version 0.1.0, written in MATLAB and running also in the free GNU Octave, so the end-user needs no licence: the certified alarms, the warnings between samples, the learned inland boundary, recharge from heads, variable-density flow and salt checked on the Henry benchmark, and the goal-oriented design of the monitoring network and of recharge from the Fushë-Kuqe work, without any field data; 36 tests); the NPD adds the model, the fast copy and the network design as the project advances.

Risks and mitigation: (1) access to data: the project drills its own boreholes on public land, and access agreements with the owners of existing wells by Month 3; at the port and the landfall the project relies only on satellite and public data; (2) security of critical-infrastructure information: every output is classified before release, asset-level vulnerability, thresholds and sensor locations stay with the end-user and the competent national authorities, and open products are aggregated or anonymised; (3) privacy: personal data are limited to workshop and training participants, collected with informed consent, minimised and pseudonymised under Albania's Law No. 124/2024 on personal data protection, aligned with the GDPR; (4) data quality: quality flags, loggers calibrated against laboratory chloride, and record errors carried as uncertainty in the model.

Work-plan

Sequencing: Phase 1, observe and measure (Months 1–12); Phase 2, model and certify (Months 4–30); Phase 3, operate, verify and hand over (Months 22–36). Gates MS3 (Month 18), MS4 (Month 22) and MS5 (Month 27) decide whether each next step starts. Every guarantee is checked on simulated test sets and on held-out field data before it is used.

WP1, Coastal motion and relative sea level (Months 1–36; NATO-country team). Steps: Sentinel-1 InSAR time series since 2014 (T1.1); corner reflectors and GNSS at the assets (T1.2); trajectory model including the 2019 earthquake and the relative sea level at each asset (T1.3); elastic and inelastic compaction and the head limit for pumping (T1.4). Risk: loss of radar coherence on soft ground; mitigation: corner reflectors and scatterers on structures. Milestone: MS2 (Month 12).

WP2, One coupled model and its certified fast copy (Months 4–24; NATO-country team). Steps: model set-up for the two sites, the Durrës wellfields and port and the TAP landfall near Fier, on one coastal motion field (T2.1); coupling of flow, salt, compaction and liquefaction, with the pore pressure of the day and the earthquake step of each layer from its shear modulus (T2.2); calibration (T2.3): the rain reaches the aquifer through a stochastic soil-water and vadose-delay model learned from irregular head records; the inputs and boundaries nobody measures (recharge coefficient, vadose delay, storage, sea level, inland inflow, initial head, dispersion width) are reconstructed by a function-space diffusion prior with an ensemble update through the exact solver; the errors of the rain product are simulated in the prior; bands are calibrated by conformal risk control on simulated aquifers and checked on measured anchor periods; the inland boundary, where the model is cut, is a conductance plus positive storage compartments on a fixed grid of time scales, passive by construction, learned from two piezometers between the wellfields and the cut, with a prior learned from simulated geology; neural-operator surrogate with a certified error bound (T2.4). Risk: the model fails validation; mitigation: build in stages and fall back to two-dimensional sections with a wider error bound. Risk: the anchors are too few (at least 9 are needed for an average guarantee, 22 for a 90% certificate); mitigation: anchors from the record since 2014 and from the new loggers, and a wider simulation-calibrated band reported as such. Risk: the inland boundary is not identified (pumping too steady, or an unknown source near the cut); mitigation: the changes of pumping in the abstraction records where they are shared, the prior from simulated geology, and a general-head boundary with a widened band as fallback. Milestones: MS3 (Month 18), MS4 (Month 22).

WP3, Certified multi-hazard alarm (Months 13–34; NATO-country team). Steps: thresholds and tolerable miss rates with the end-user, the Institute of Geosciences, IGJEO (T3.1); alarm margins and proofs (T3.2), with anchor calibration so that the guarantee does not depend on the simulator, horizon matching under drift, and the floor that shows how far each band is from the narrowest possible, and warnings for crossings between samples: the chance that chloride crosses its limit at any time in the season, as a band over all models that fit the records, widened by a conformal margin calibrated on simulated aquifers and anchor periods, and issued online as the conditional probability given the irregular samples and the daily heads; joint guarantee and allocation of tolerances (T3.3); back-testing (T3.4); verification in operation (T3.5). Risk: too many false alarms; mitigation: acceptable rates agreed in advance and tolerances allocated by consequence. Risk: drift faster than the record allows; mitigation: report the longest horizon for which a band exists. Risk: chloride more skewed than the model family; mitigation: the conformal margin, re-estimated each season. Milestone: MS5 (Month 27).

WP4, Monitoring network designed for the warnings (Months 1–36; NATO-country team). Steps: design (T4.1), in which each candidate sensor is valued by the guaranteed band it removes, its floor and the years of safe operation it adds per euro, with a local rain gauge among the candidates and one logger next to the wellfield most exposed to local demand; the candidate sites are boreholes on public land and existing wells whose owners agree; a barometer and a sealed reference channel on each logger hub are candidates too, the error model carries the barometric efficiency of each well, the drift of each logger, cable cross-talk and model error within stated ranges, and the network is chosen against the worst plausible error and reported with its certified optimality gap; a conductivity logger in the monitoring well closest to each wellfield and a yearly campaign of laboratory samples paired 1, 3 and 7 days apart measure the time scale that decides crossings between samples; a tracer campaign (Br/Cl, water isotopes, depth profiles from multilevel wells, nitrate and boron) is chosen by the value of the pumping decisions it changes; two piezometers between the wellfields and the inland cut, about one diffusion length apart, learn the boundary of the model (T2.3); procurement and installation (T4.2): drilling of the boreholes and loggers in the chosen wells; near the port and the TAP landfall, corner reflectors, GNSS and a seismic station on public land; operation and maintenance (T4.3), in which every reading is corrected for air pressure and logger drift before use and the corrected residuals are checked for remaining correlation; seismic velocity as a water-table gauge, and the shear-wave velocity of the soft layers for their earthquake step (T4.4). Risk: procurement delays and limited site access; mitigation: early orders; satellite and public data as the baseline at the port and the landfall. Risk: lagged or uncertain barometric responses; mitigation: response functions estimated from the first three months by

regression deconvolution, with their error carried into the robust design. Milestones: MS1 (Month 6), MS2 (Month 12).

WP5, Live demonstration (Months 22–36; NATO-country team). Steps: dry run (T5.1); twelve months of bulletins (T5.2); safe operating window and pumping rule (T5.3): the effect of each wellfield's pumping on salinity is estimated with the hidden drought demand removed, by reading the recharge from the heads after adding back the drawdown estimated from the abstraction records where they are shared (otherwise from the heads), and a calibrated check of every substitute is run before any effect is reported; the salinity under a drier recharge regime is computed from the calibrated state with conformal bands; the certified window, its closing year and a plan of safe pumping tests that would sharpen it follow, and the pumping and drought rule is prepared for the authorities that issue water-use permits; while the source of the chloride is uncertain (sea, deep brine, drought or return flow), the sources the tracers cannot exclude are kept in a set updated monthly with anytime-valid e-values, the pumping allocation is kept safe for every source in it, and a ledger shows the cost borne by the water supply, the farmers and the wetland; post-event protocol (T5.4), in which the relaxation of the logged heads after a strong earthquake gives the leakage of the confined aquifer as a free pumping test. Risk: no strong event occurs; mitigation: verification by back-testing and storylines, and the well records of November 2019 where they exist. Risk: drawdown coefficients from the abstraction records are uncertain; mitigation: their uncertainty is carried into the effect as a stated bias band. Risk: the true source of the chloride is not among those considered, or mixes two of them; mitigation: a goodness-of-fit check at every update, mixtures admitted as hypotheses, and the alternative supply as fallback. Milestone: MS6 (Month 34).

WP6, Training, transfer and dissemination (Months 1–36; Qatar University team): early-career researchers and a training school (T6.1), training of the staff of IGJEO and of the intended users (T6.2), open release (T6.3), transfer to the east coast of Qatar (T6.4). Risk: users rely on the bulletins alone, or trained staff leave; mitigation: the system stays advisory beside the users' own procedures, the training explains what a certified miss rate means, and at least four staff of IGJEO are trained. Milestone: MS7 (Month 36), hand-over to IGJEO: an operating procedure for the bulletins agreed with it, the bulletins of Months 34–36 used by its trained staff with the project team only in support, and an agreement on keeping and maintaining the sensors signed by Month 33 (D6.2).

WP7, Management, data and risk (Months 1–36; coordinator): management and reporting (T7.1), data plan and access agreements (T7.2), risk register (T7.3). Risk: delays in drilling permits or in access to existing wells or records; mitigation: access agreements by Month 3, with WP1 and the design of WP4 started on public and satellite data meanwhile. Risk: a release discloses sensitive information on critical infrastructure; mitigation: data classified in the data plan, asset-level results shared only with the end-user and the competent national authorities, and publications checked for sensitive information on the assets before submission (T7.2). Milestones: access agreements and data management plan (D7.1, Month 3); progress reports at the SPS milestones (D7.2, Months 6, 18 and 30); final report (D7.3, Month 36); the risk register reviewed at every gate.

After the project ends: Sustainability and role of end-user(s)

Implementation. The Institute of Geosciences (IGJEO) of the Polytechnic University of Tirana, the end-user, which monitors earthquakes, weather and waters in Albania, intends to use the bulletins and the open methods in its monitoring. In Months 34–36 its trained staff use the bulletins with the project team only in support. The network stays in Albania with the Polytechnic University of Tirana, which keeps the bulletins running, and an agreement (Month 33) sets out how the sensors are kept and who carries the running cost of about €10,900 per year for data, telemetry and spare sensors. An operating procedure for the bulletins and a policy brief for the national authorities (D6.2) anchor the warnings in practice.

Next steps: make the warnings for the wellfields, the port and the landfall available to the national authorities through IGJEO; extend the warnings to the other wellfields and coastal assets of Albania; offer the open software to coasts of Allies and partners (tested on the east coast of Qatar during the project); seek national and European funding to keep the network and the verification ledger running.

Role of the end-user. The Institute of Geosciences (IGJEO) of the Polytechnic University of Tirana, which monitors earthquakes, weather and waters in Albania, is the end-user: it designates a contact

person and a small team, who sit on the steering committee, agree the warning thresholds for earthquakes and extreme sea levels (T3.1), review the bulletins during the live operation (T5.2) and take part in the final workshop; it shares the data of its stations under a data agreement; its staff are trained, and it receives the software and the operating procedures at the hand-over (MS7). Its contribution is staff time and data in kind, without procurement, ownership or maintenance obligations. End-users receive no SPS funds.

Risks and mitigation strategies

Risk description	Likelihood	Impact	Mitigating action
Drilling permits or access to existing wells and records are delayed; calibration and design start late	Medium	High	The project drills its own boreholes on public land; permits requested in Month 1; access agreements with the owners of existing wells by Month 3 (T7.2, D7.1); the port and the landfall rely only on satellite and public data
Historical head and chloride records are incomplete or of uneven quality	High	Medium	Data rescue from paper archives; quality flags; record errors carried as uncertainty in the assimilation
The coupled model fails its validation test on held-out data	Medium	High	Build in stages (flow, then salt, then compaction and liquefaction); test each against data before coupling
Certified alarms are too cautious: the false-alarm rate is unacceptable to the users	Medium	High	Acceptable false-alarm rate agreed with the end-user (the Institute of Geosciences, IGJEO) at T3.1; tolerances allocated by consequence; network designed to shrink margins
Conditions move beyond the observed record (drought, compound storm, post-seismic change) and weaken the guarantees	Medium	High	Extreme-aware margin term; storylines beyond the record in back-testing (T3.4); nonstationary extremes with land motion
No strong earthquake or surge occurs during the demonstration, so the guarantees are not tested on such events	High	Medium	Back-testing over the historical record, including the 2019 earthquake, and on storylines beyond it (T3.4); exceedance events per alarm counted first, since a 10% miss rate needs at least 29 events to verify (Methodology 5.7)
A missed alarm coincides with damage; trust and responsibilities are questioned	Low	High	Every bulletin states its certified miss rate; decisions stay with the users; roles written into the bulletin protocol (T5.1)
Sensitive information on critical-infrastructure vulnerability is disclosed through open data or publications	Medium	High	Data classification in the data plan (T7.2); asset-level results shared only with the end-user and the competent national authorities; open release of methods, code and non-sensitive aggregated data; publications are checked for sensitive information on the assets before submission

All 22 risks, with owners and triggers, are in the Risk Assessment document.

Communication Plan

Activities: a public kick-off, a public start of live operation and a final event; a project website with public summaries in Albanian and English; a quarterly verification ledger showing, for each alarm type, events, misses, false alarms and the bound on the miss rate; at least six open-access papers and eight conference presentations; a policy brief for the national authorities; stakeholder workshops at kick-off, mid-term, end and for transfer.

Presenting the project to the public: the start of live warnings, the training school (Month 14) and the hand-over to IGJEO are natural moments for press releases and local media.

Likely photo and video subjects for the SPS Office (after a check for sensitive information on the assets): installing piezometers and GNSS stations, corner reflectors on the coast, the training school, a bulletin exercise with IGJEO.

Budget link: the heading "Communication and Publications" (€8,000) covers open-access fees (€6,000), the website, translation and printing (€1,500) and public events (€500); further open-access papers go to diamond open-access journals or to Zenodo. Asset-level information is never published; every output passes a release gate.

Key Performance Indicators (KPIs)

500 words.

The project is on track when the following are achieved on time:

- D4.1 network design approved by the steering committee, with a certified optimality gap of at most 25% for the chosen network (Month 6, MS1).
- D4.2 network installed and streaming with at least 90% data availability, and the corrected heads within 3 cm root-mean-square of manual dips (at most half the raw error); D1.1 coastal motion atlas with a vertical velocity uncertainty of about 1–2 mm/yr at asset points (Month 12, MS2).
- D2.1 calibrated coupled model: held-out heads, chloride, InSAR and seismic velocity inside the 90% predictive intervals at close to the nominal rate, and for each wellfield a certified band at most 0.25 wide for the chance that chloride crosses its limit within the season (Month 18, MS3).
- D2.2 certified surrogate: error bound met on test runs; alarm prototype passes a dry run (Month 22, MS4, go/no-go for live operation).
- D3.2 back-testing: each alarm's miss rate at or below its target with a 95% confidence bound, crossings between samples included, under a protocol published with a DOI before the results (Month 27, MS5).
- D5.1 twelve months of live bulletins with every outcome logged, the miss rates consistent with their certificates, and the certified safe operating window reported for each asset, with a plan of safe pumping tests and the years it would add (Month 34, MS6).
- Source of the chloride (T5.3): the tracer campaign chosen by decision value completed by Month 24, and the set of plausible sources with the pumping allocation published monthly from then on, keeping the true source in at least 95% of back-tested simulated cases.
- Inland boundary (T2.3): learned from the two boundary piezometers and passive (all weights nonnegative), with held-out heads at both piezometers inside the 90% predictive intervals (Month 18, MS3).
- Earthquake response (T4.4, T5.4): the shear-wave velocity of the soft layers near the port and the landfall measured by Month 18, giving the step coefficient of each layer; after any earthquake felt at the sites, the leakage of the confined aquifer read from the loggers within 30 days.
- D6.1 open software and data with a DOI, reproduced by someone outside the team (Month 33); D6.2 operating procedure agreed with IGJEO and policy brief delivered (Month 36, MS7).
- Capacity: at least 15 trained professionals (four from IGJEO), at least six early-career researchers, at least six open-access papers and eight conference presentations.
- Management: progress reports at each SPS milestone, spending within 10% of the plan per budget category, and the risk register reviewed at every gate.

435 words (limit 500)

Criteria for success

At most seven, SMART, weights summing to 100%.

Criterion	Relative weight (%)
Each alarm's miss rate in back-testing over the record since 2014 is at or below its certified target (indicatively 5–10%), with a 95% confidence bound (D3.2, Month 27)	25
Twelve months of live bulletins (Months 23–34) with every outcome logged and no verification test rejecting a certified miss rate (D5.1)	20
Held-out heads, chloride, InSAR and seismic velocity fall inside the model's 90% predictive intervals at a rate of at least 85% (D2.1, Month 18)	15
Trained staff of IGJEO use the bulletins in Months 34–36 with the project team only in support, and the agreement on keeping the sensors is signed (MS7)	15
The monitoring network reaches at least 90% data availability over twelve months of operation (D4.2, T4.3)	10
The certified safe operating window and its closing year are reported for each asset and confirmed by the full model at the closing point, with a plan of safe pumping tests and the years it would add (T5.3)	10
The open release is installed and reproduces the synthetic test coast for a group outside the consortium (D6.1, Month 33)	5

List of abbreviations

AI: artificial intelligence; CF: Climate and Forecast conventions; CPT: cone penetration test; D: deliverable; DOI: digital object identifier; dv/v: relative change of seismic velocity; EDT: emerging and disruptive technologies; FAIR: findable, accessible, interoperable, reusable; GDPR: General Data Protection Regulation; GNSS: global navigation satellite system; InSAR: interferometric synthetic aperture radar; KPI: key performance indicator; MS: milestone; MYP: Multi-Year Project; NPD: NATO-country Project Director; PPD: partner-country Project Director; SPS: Science for Peace and Security; T: task; TRL: technology readiness level; WP: work package.

Optional upload: a file that helps to illustrate the project

Upload CERTAIN-COAST_Illustrative_Figures.pdf (ready): six pages, a table of contents and twelve figures (the coast as one system, the critical assets and NATO resilience, the method, the innovations in the warning chain, the gaps closed, the certified safe operating window, single-hazard horizons, the alarm margin, warnings between samples, the self-correcting network, milestones and gates, impact pathway).

8. Project Schedule & Budget

Project duration in months

36

Project schedule (upload)

PDF with a Gantt chart: duration and interdependence of each step, critical milestones, deliverables.

Upload CERTAIN-COAST_Project_Schedule.pdf (ready): Gantt chart with the duration of every task, the dependencies, the critical path, the milestones and the deliverables by SPS milestone. Project milestones: MS1 network design (Month 6), MS2 network streaming (Month 12), MS3 calibrated model (Month 18), MS4 go/no-go for live operation (Month 22), MS5 miss rates confirmed (Month 27), MS6 twelve months live (Month 34), MS7 hand-over (Month 36).

Milestones overview

SPS reporting milestones; must match the "Info" tab of the budget workbook (at least one milestone every 12 months).

Event	Description	Months after kick-off
Kick-off	Project kick-off meeting	0
Milestone 1 (required)	First technical and financial report	6
Milestone 2 (required)	Second technical and financial report	18
Milestone 3 (required)	Third technical and financial report	30
Milestone 4 (optional)	Fourth technical and financial report	leave blank
Final report (required)	Final technical and financial report	36

Budget summary by item: NATO SPS funding requested (in EUR)

Indicative split by reporting period, assuming even spending within each project year (period ending at Milestone 1 = months 0–6, M2 = 6–18, M3 = 18–30, final = 30–36). Enter the final figures from your Excel workbook.

Budget category	M1	M2	M3	M4 (opt.)	Final	Total
Equipment	75,900	91,400	15,500	—	0	182,800
Training	2,100	14,800	18,400	—	5,700	41,000
Communication and Publications	363	1,575	3,637	—	2,425	8,000
Travel	6,900	14,700	17,100	—	9,300	48,000
Consumables	3,805	7,878	7,545	—	3,472	22,700
Other	6,000	10,500	10,000	—	5,500	32,000
Stipends	7,200	16,800	16,800	—	7,200	48,000
Co-financing of personnel	0	0	0	—	0	0
Total	**102,268**	**157,653**	**88,982**	**—**	**33,597**	**382,500**

Equipment

Major equipment, its role, approximate price, co-funding.

- Monitoring boreholes: drilling, casing, screens, sealing: €32,000. Piezometers where the network design (D4.1) ranks them highest; sealed to standard (R12)

- Pressure, conductivity and temperature loggers: €14,400. Heads and salinity in new and existing wells: the core data for intrusion and the water table
- Barometric loggers: €1,200. Barometric correction of the pressure loggers
- Telemetry units (mobile data loggers): €7,200. Near-real-time data for the live bulletins (WP5)
- GNSS stations: receiver, antenna, monument: €27,000. Vertical land motion at the project's boreholes near Durrës and on public land near the port and the TAP landfall; ties InSAR to a global frame (T1.2)
- InSAR corner reflectors: €9,000. Stable radar targets on soft ground at the assets (R4)
- Seismic stations: three-component sensor and digitiser: €30,000. Continuous ambient noise for seismic velocity along the coast near the port of Durrës and the TAP landfall (T4.4)
- Seismic node rental, two campaigns: €8,000. Dense temporary coverage to calibrate velocity against the water table
- Fibre-optic sensing campaign: rental and processing (conditional): €15,000. Released only if fibre access is agreed by Month 9 (R9); otherwise moved to seismic nodes
- Power and enclosures: solar panels, batteries, locked boxes: €7,000. Autonomous and secure operation of the field stations (R11)
- Field chemistry instruments: multiparameter meter, sampling pump: €6,000. Conductivity and temperature in the field; sampling for chloride
- Spare loggers and sensors: €6,000. Replacement within two weeks after damage or theft (R11, R21)
- Computing server with GPU: €12,000. Thousands of coupled-model runs and training of the certified surrogate (T2.4)
- Redundant data server and backup storage: €5,000. Operational redundancy for the live bulletins (R17)
- Software licences: €3,000. Processing tools without an open-source equivalent; the project itself releases open software

Co-funding: [existing equipment and facilities of the institutions]; the Institute of Geosciences (IGJEO), the end-user, contributes staff time and station data in kind.

Training

SPS does not fund fellowships; training periods are short.

- Young-researcher exchanges, two months each: €21,000. Partner-country early-career researchers work on site with the data and the end-user (T6.1)
- Training school: venue, materials, trainee travel: €15,000. Hands-on school with the coast kit in Month 14 (T6.1)
- Training sessions for the end-user's staff and intended users: €5,000. Staff able to run the system unaided (T6.2)

Travel

- Consortium meetings (kick-off, mid-term, final): travel and subsistence: €21,600. Six people from the two countries at three meetings
- Stakeholder workshops: kick-off, mid-term, final, transfer: €12,000. Thresholds, dry run, hand-over and transfer to other coasts (Dissemination Plan)
- Conference presentations: €14,400. At least eight presentations of the results (Dissemination Plan)

Co-financing of personnel

Up to 25% of the personnel cost.

None requested: the institutions pay all salaries (see Non-NATO funding). The management hours of the NPD (€15,000) are under "Other", as the MYP Handbook (April 2026) requires. [If salary co-financing is wanted after all: per person and month at most the share of time on the project, capped at 25%, times the monthly gross salary; in total up to about one third of the grant; a CV for each recipient.]

Other major expenses (optional)

• Open-source software of the model and the alarm algorithm: code repository, continuous integration, cloud computing for tests and releases, archiving with a DOI: €10,000. The NPD builds the coupled model and the alarm algorithm and releases them as open-source code for the users (D6.1)

• Shipping and customs for equipment: €4,000. Delivery of sensors to the site

• Insurance of field equipment: €3,000. One year of cover per year of operation

• Man-hours for the management of the project (NATO-country Project Director only): €15,000. Coordination, reporting and financial records, with time records; SPS category "Other"; about 4% of the total

Communication and Publications (€8,000) and Consumables (€22,700) are detailed in the budget workbook and the Communication Plan above.

Budget summary by item: non-NATO funding contribution (in EUR)

Estimate what the institutions and nations contribute.

Budget category	What it covers	Amount from other funding sources
Equipment	Existing equipment used by the project: field instruments, laboratory instruments and servers of the institutions; existing wells whose owners agree to their use; existing seismic stations. Value: [share of the replacement value or depreciation over the 36 months].	[€ amount]
Training	Venues, rooms and trainers' time of the institutions for the training school and the training of the end-user's staff; the trainees' staff time is under "Other".	[€ amount]
Communication & Publication	[Institutional open-access agreements or press offices, if any; otherwise 0].	[€ amount]
Travel	[Institutional travel funds used for the project, if any; otherwise 0].	[€ amount]
Consumables	[Laboratory chloride and major-ion analyses done by an institution laboratory at its own cost, if agreed; otherwise 0].	[€ amount]
Stipends	None expected (stipends of the early-career researchers are requested from SPS).	[€ amount]
Co-financing	Personnel cost paid by the institutions: all salaries of the Co-Directors, researchers and staff for their time on the project (Section 3). SPS co-finances no salary in this version.	[€ amount]
Other	Overheads of the institutions (offices, administration, IT) at their own rates; staff time and station data of the end-user (the Institute of Geosciences, IGJEO); data provided under agreements (Durrës tide-gauge record of ASIG, records of the national seismic stations, records of existing wells shared by their owners).	[€ amount]

Amounts, salaries and the personnel worksheet: CERTAIN-COAST_Non-NATO_Contribution.docx.

Non-NATO funding

The SPS grant pays only project-specific costs. The participating institutions and the end-users contribute the rest:

• Salaries of the Co-Directors, researchers and staff, paid by their institutions for about 66.6 person-months over 36 months (NPD 21.6, PPD 7.2, hydrogeologist of the Faculty of Geology and Mining 10.8, two researchers and a laboratory technician of the Faculty 18.0, financial officer of UPT 3.6, three researchers of Qatar University 5.4): [€ amount]. No salary co-financing is requested from SPS.

• Overheads of the institutions (offices, laboratories, IT, administration) at their own rates: [€ amount].

• Existing equipment and facilities used by the project: [field and laboratory instruments, servers] of the institutions: [€ amount].

• The end-user, which receives no SPS funds: staff time of the Institute of Geosciences (IGJEO) for the steering committee, thresholds, the dry run, the live bulletins and training (about 1.8 person-months): [€ amount].

• Data at no cost to the project: Copernicus Sentinel-1 radar images (free and open), the Durrës tide-gauge record of ASIG and the records of the national seismic stations under data agreements [not valued / € amount].

• [Other national or institutional co-funding, only if confirmed: source and amount].

Total non-NATO contribution: [€ amount], [x]% of the SPS grant (€382,500).

How do you want to share the financial rules of the NPD's institution?

Choose "Link to a website": https://upt.edu.al/wp-content/uploads/2023/02/Statuti-PDF-English.pdf (Statute of the Polytechnic University of Tirana, English version published by UPT: Chapter IV "Properties, financing and services", Articles 24–25; Article 10, the Board of Administration approves the financial regulation). The UPT finance office confirmed on 10 November 2026 that UPT has no separate internal financial regulation: its funds, including those of foreign-funded projects, are administered under Law no. 80/2015 on higher education and scientific research, Law no. 9936/2008 on the management of the budget system, the annual budget law, Law no. 25/2018 on accounting and financial statements, Instruction no. 9 of 20.03.2018 of the Minister of Finance and Economy on standard procedures for budget execution, Instruction no. 10 of 12.07.2024 of the Ministry of Finance on the registration, use, reconciliation, reporting and publication of foreign funds under international agreements, and Instruction no. 30 of 27.12.2011 on the management of assets in public sector units. It also confirmed that, under Instruction no. 10/2024, the funds of a project are managed in the revenue account of UPT at the State Treasury Branch and used only for the purpose approved in the financing agreement.

9. End-Users

At least one end-user; all based in a NATO or eligible partner country; end-users receive no SPS funds.

First name	Family name	Title	Job title	Institution	Postal address	Country	Telephone	E-mail
Ylber	Muceku	Prof. Dr.	[Job title]	Institute of Geosciences (IGJEO), Polytechnic University of Tirana	Rruga Don Bosko 60, Tirana	Albania	[Telephone]	[E-mail]

End-User(s) commitments (upload)

One PDF with the signed letter of support and end-user commitment of the Institute of Geosciences (IGJEO) of the Polytechnic University of Tirana, on its letterhead, signed by Prof. Dr. Ylber Muceku (draft: CERTAIN-COAST_Letter_IGJEO_EN_SQ.docx, page 1 in English to sign; page 2 is an Albanian reading copy). Add the letter of any other end-user that signs before the deadline. It cannot follow after the deadline: the call and the MYP Handbook list the end-user commitment among the documents submitted with the proposal; only the IPR agreement may follow approval.

10. Project attachments

Brief overview of participating institutions (upload)

One PDF, at most one page per institution: capabilities, resources, facilities, websites. Use CERTAIN-COAST_Institutions_Overview.docx: [fill the yellow fields, delete the pages of institutions that do not take part], then save it as PDF.

SPS MYP Agreement for Joint Research (upload)

NATO template, signed by all Co-Directors and a representative of each institution.

Do you want to include the Intellectual Property Rights (IPR) Agreement (or Waiver) now?

Yes (recommended): upload the signed agreement as one PDF. Draft: CERTAIN-COAST_IPR_Agreement_Draft.docx (NATO sample, Option B, open release of software and data, security release check), with a one-page waiver as alternative. The form and the Handbook allow it after approval, but the 2027-1 call lists it among the documents of a complete proposal.

Additional files (optional)

Upload CERTAIN-COAST_Additional_Files.pdf (92 pages), one PDF with a cover and a table of contents at the start, as the form asks: the state of the art (15 pages: what 30 key papers of 2020–2026 from Caltech, MIT, Harvard and Oxford have done, what they have not done, and what CERTAIN-

COAST brings in mathematics and physics), the mathematical state of the art (7 pages: random forcing, boundaries, initial state and propagation of uncertainty, and the four numbers that choose the method), the technical annex (10 pages: how the methods work, the preparatory tests, the gates with their fallbacks and the nine new contributions, consistent with this form), the technical note on the assembled system of equations and the placement of recharge and supply wells (23 pages; synthetic tests, illustrative values), the BEYOND-GO theory note on the inland boundary of the model (16 pages; four propositions with proofs, synthetic tests), the SHOCK-GO implementation report on the earthquake in the aquifer (14 pages; code-based checks, synthetic tests) and the CVs of the NATO-country and partner-country Project Directors (2 pages each).

Additional information (optional)

The additional files open with the state of the art, what 30 key papers of 2020–2026 have done, what they have not done and what CERTAIN-COAST brings, and with a mathematical state of the art. They continue with the technical annex, which summarises the methods and their preparatory tests, a technical note on the coupled model and the placement of recharge and supply wells, a theory note on the inland boundary of that model, learned from piezometers, and an implementation report on the earthquake in the aquifer, and close with the CVs of the two Project Directors; the full methodology, work packages, risk assessment and data management plan are available on request.